

# OLUBUNMI AKINREMI

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Lausanne, Switzerland

## TOP ACHIEVEMENTS

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- **NASA Space Apps Hackathon:** Global Finalist
- £70,000 Research Grant Award from the French Embassy Fund
- **2025 Top 30** Top 30 African AI Researchers feature by OneDev
- Innovative Data Guru feature by Tech Trend Africa
- **Miss Algorithm Award** at DSN Bootcamp

## WORK EXPERIENCE

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- **Laboratory for Intelligent Global Health & Humanitarian Response Technologies** *EPFL, Switzerland*  
*Research Fellowship* June 2026 - August 2026
  - Developed an assistive Ebola triage tool and severity monitoring tool, aimed at 5% reduction in death by early detection of the virus in current pandemic areas like DR Congo.
  - trueMOOVE: led the development an Agentic AI feedback system as part of a clinical diagnostic tool to detect missing information in voice dictated medical notes in 5 African languages.
- **BFree Africa** *February 2026 - May 2026*  
*AI Engineer* Lagos, Nigeria
  - **End to End AI Call Agent:** Architected and shipped a production AI voice agent for loan recovery, integrating FastAPI, FastRTC/Twilio Media Streams, in house finetuned Whisper STT and XTTSv2 (AfroTTS Nigerian voice), and LangChain into a single real-time telephony pipeline.
  - Designed a 5-stage conversational state machine with per-stage tool-access controls, persona prompts, and a structured summary that preserves agreed amounts, dates, and objections across the full call.
  - Built a two-tier intent classifier combining an Aho-Corasick keyword automaton (O(n)) with an LLM fallback.
  - Implemented early barge-in handling, DTLN noise suppression, and filler caching to meet a <1.5 s average conversational gap target.
  - Fine-tuned Whisper (LoRA adapter on AfriSpeech corpus) with stage-aware decoder prompts to reduce domain homophones; benchmarked and optimised STT/VAD parameters via Optuna.
  - Delivered CI/CD pipeline, per-turn latency observability, and a companion ml-hub monitoring dashboard providing real-time STT/LLM/TTS latency KPIs and Slack alerting.
- **Kochava** *March 2023 - January 2026*  
*Machine Learning Engineer* Idaho, United States
  - Slashed model training runtime by 50% by engineering a distributed training system on Kubernetes with MLFlow, resolving critical CPU bottlenecks and enabling concurrent model training.
  - Improved budget spend forecast accuracy by 50% and cut modelling time by 30% by improving legacy statistical models using fine grained mathematical and analytical techniques.
  - Achieved 95% accuracy in data quality assurance by developing an automated anomaly detection system to smooth and interpolate raw client data.
  - Reduced client onboarding time to budget spend forecast by 40% by architecting a Kubeflow pipeline that automatically transforms data science notebooks into production-ready, SOLID-compliant code.
- **Centre for Journalism and Innovation Development(CJID)** *September 2022 - August 2023, April 2024 - August 2024*  
*Machine Learning Engineer (GenAI & LLMs)* Abuja, Nigeria
  - Spearheaded the organisation's first GenAI-powered platform, automating audio transcription for local languages and reducing manual labour for journalists by 70%.
  - Fine-tuned and evaluated open-source LLMs (BERT, Whisper) on domain-specific African datasets to enhance fake news detection capabilities.

- Constructed robust data ingestion and preprocessing pipelines for automated audio clipping, annotation, and metadata generation.

## • Rural Farmers Hub

Machine Learning Engineer

September 2022 - February 2023

Abuja, Nigeria

- Reduced GIS analysis time by 25% by optimising satellite image quality using OpenCV tools.
- Eliminated dependency on third-party GIS tools and improved inference accuracy by 20% by deploying a custom ML-based real-time Soil Organic Matter (SOM) estimator.

## PROJECTS

### • LLM-enabled Agent Communication Bots

2024 - Present

Tools: GitHub Actions, Langchain, GSpread, Azure, BeautifulSoup



- [Multimodal LLM Recipe Recommendation Engine](#)
- [RAG-based educational content platform](#)
- [E-commerce daily product recommendation agent](#)

### • Reinforcement Learning Agents

Dec 2022 - August 2026

PyBullet, Stable-Baselines3, PPO, PyTorch



- [Sea-X](#): Finedtuned computer vision models to detect waste types, observed through UAV aerial shots, common to African river bodies.
- [Adaptive RL for Drone Navigation](#). Trained a PPO (GPU-backed) policy network on a custom environment with a 35-dimensional observation space and reward function balancing waste patch discovery(using the trained model above), obstacle avoidance, and step efficiency.
- [MARL-Gym: Open-Source Multi-Agent RL Platform](#). A monaco based full-stack web platform for training and evaluating RL and multi-agent RL systems, featuring environment selection from Gymnasium and PettingZoo

## EDUCATION

### • Obafemi Awolowo University

April 2019 - August 2025

B.Sc. Computer Science with Mathematics | GPA: 4.4/5.0

Ile-Ife, Nigeria

- **Financial Fraud Detection System**: Led a research project investigating fraud patterns in financial institutions, utilising statistical analysis to identify anomalies.
- **Robotic Arm Control**: Spearheaded the Python implementation for a robotic arm control system for the Introduction to Artificial Intelligence course.

### • Massachusetts Institute of Technology (MITx)

August 2023 - August 2024

MicroMasters® Program in Data Economics and Development Policy

Online

Status: Graduate-Level Credential

## PUBLICATIONS & POSTERS C=CONFERENCE, P=POSTER

### 4.1 Edited and Referred Conference Proceedings:

- [C.1] ["Emezue, C., Community, N., Awobade, B., Owodunni, A., Emezue, H., Emezue, G.M.T., Emezue, N.N., Ogun, S., Akinremi, B., Adelani, D.I. and Pal, C., 2025. The NaijaVoices Dataset: Cultivating Large-Scale, High-Quality, Culturally-Rich Speech Data for African Languages."](#) arXiv preprint arXiv:2505.20564 (2025).

### 4.2 Submitted for Review:

- [C.2] "Samuel Ibiyemi, Bunmi Akinremi, Jennifer Ilesan. Overcoming Degradation in African Low Cost Locomotive Sensors using RL." Special Issue, African Cities Journal (2026).
- [C.2] "Bunmi Akinremi, Ishaya Gambo, Rhodes Massenon, Samuel Okegbile, Neha Agarwal, Saurabh Agarwal. Autonomous Plastic Waste Patch Detection in Aquatic Environments Using Deep Reinforcement Learning with Visual Feedback." International Journal of Computer Vision, Springer (2025).

### 4.3 Non-Referred:

- [C.3] Bunmi Akinremi, Fajemila Dunsin, Paul Okewunmi, Victor Olufemi (2024). [Quantization Techniques for Diacritized Yoruba Transformer Models](#) In *African NLP workshop*
- [P.1] Bunmi Akinremi, Paul Okewunmi (2023). [Unmasking "Fake News" For Fact-Checking Radio Shows via AI-Driven Claim Extraction](#). Poster Presentation at *Deep Learning Indaba 2023*.
- [P.2] Bunmi Akinremi (2022).[Detection and Analysis of Plastic Waste: A Tiling Approach](#). Poster Presentation at *Deep Learning Indaba 2022*.